

Assignment: Polynomial Regression

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Target Metric: Test Generalization
(MSE & R^2)

Executive Summary & Key Findings

- **Problem 1 (var1 — Steam Turbine Optimization):** Optimal model is **Degree 5 Sparse Regularized Polynomial Regression** across all 6 operational parameters. While naive unregularized OLS peaked at degree 4 (CV $R^2 = 0.9043$, MSE = 0.9618) due to severe variance inflation from 462 terms, applying **L1 Lasso screening** ($\alpha = 0.018$) + **Post-Lasso OLS debiased refitting** ($t > 1.4$) isolates **47 active interaction terms**, achieving **5-Fold CV MSE = 0.2913 ± 0.0248** and **CV $R^2 = 0.9711 \pm 0.0031$** (a **69.7% error reduction**), matching the irreducible sensor noise floor ($\sigma \approx 0.51$).
- **Problem 2 (var2 — Subterranean Thermal Mapping):** Optimal model is **Degree 8** using **all 3 spatial coordinates** (165 terms, OLS), achieving **5-Fold CV MSE = 0.2523 ± 0.0275** and **CV $R^2 = 0.9946 \pm 0.0008$** . Degree 8 resolves the complete 3D thermal convective field down to the physical sensor noise floor ($\sigma^2 \approx 0.20$ – 0.25).

Abstract — This report presents a rigorous empirical investigation into polynomial regression for two high-impact geothermal energy engineering domains: calibrating surface turbine thermodynamic power output (Phase 1, var1) and mapping subterranean 3D geothermal temperature anomaly structures (Phase 2, var2). Leveraging stratified 5-fold cross-validation protocols within *scikit-learn*, we evaluate exhaustive degree sweeps, combinatorial interaction manifolds, and regularized basis selection. For Phase 1, we uncover the sparse degree-5 polynomial structure governing turbine efficiency, raising cross-validation R^2 from 0.9043 to 0.9711. For Phase 2, full degree-8 polynomial regression achieves $R^2 = 0.9946$, perfectly capturing complex 3D thermal harmonics.

1. Introduction and Problem Formulations

Polynomial regression provides a flexible linear-in-the-parameters framework capable of capturing complex non-linear response surfaces. Given raw input vector $\mathbf{x} \in \mathbb{R}^D$, polynomial expansion synthesizes cross-feature multiplicative and power interactions. This study addresses two distinct geothermal engineering phases under personalized calibration data:

- **Phase 1: Power Plant Steam Turbine Optimization (var1):** Optimizing surface electricity generation before well drilling. The turbine system is governed by six continuous control parameters normalized in $[-1.0, 1.0]$: high-pressure steam valve adjustment (x_1), condenser coolant flow rate (x_2), re-injection pump pressure (x_3), turbine blade pitch angle (x_4), exhaust valve rate (x_5), and steam inlet pressure (x_6). The target variable y is the Net Power Score (training mean 0.8629, std 3.1872, range $[-9.879, 12.067]$). The domain is characterized by moderate-degree non-linearities (up to degree 10).
- **Phase 2: Subterranean Thermal Reservoir Mapping (var2):** Identifying optimal extraction drilling targets from seismic and thermal probe sensors. Inputs represent 3D spatial coordinate offsets from basecamp: East-West offset (x_1), North-South offset (x_2), and depth offset (x_3) bounded in $[-1.0, 1.0]$. The target y is the Thermal Anomaly Score (mean 2.2751, std 6.9016, range $[-30.257, 39.328]$). The convective subsurface temperature field exhibits steep spatial gradients requiring higher-order polynomial harmonics (up to degree 20).

Dataset	Features	Train N	Test N	Target (y) Mean $\pm$ Std	Target Min / Max
IMT2024072_var1	6 ($x_1 \dots x_6$)	1,000	1,000	0.8629 ± 3.1872	-9.879 / +12.067
IMT2024072_var2	3 (x_1, x_2, x_3)	1,000	1,000	2.2751 ± 6.9016	-30.257 / +39.328

Table 1: Summary of calibrated geothermal datasets for Roll Number IMT2024072.

2. Mathematical Formulation and Validation Protocol

For an input vector $\mathbf{x} = [x_1, \dots, x_D]^T \in \mathbb{R}^D$, polynomial regression of degree d maps the inputs into a linear combination of basis monomial functions:

$$y = w_0 + \sum_{i=1}^D w_i x_i + \sum_{i=1}^D \sum_{j=i}^D w_{ij} x_i x_j + \dots = \Phi(\mathbf{x})^T \mathbf{w} + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2) \quad (1)$$

where $\Phi(\mathbf{x})$ contains all monomial terms whose power sum satisfies $\sum_{k=1}^D p_k \leq d$. The parameter count scales combinatorially according to the multi-index binomial coefficient:

$$P = C(D + d, d) = (D + d)! / (D! d!) \quad (2)$$

For Phase 1 ($D = 6$), P grows aggressively: degree 1 has 7 terms, degree 2 has 28, degree 3 has 84, degree 4 has 210, degree 5 has 462, and degree 6 has 924 terms. For Phase 2 ($D = 3$), scaling is compact: degree 4 has 35 terms, degree 6 has 84, degree 8 has 165, and degree 10 has 286 terms.

Cross-Validation Protocol & Regularization: Models were evaluated using **5-Fold Cross-Validation** (*KFold*, *shuffle=True*, *random_state=42*). In Phase 1, unconstrained OLS over-fits at degree 5 ($P = 462$ on $N_{\text{train}} = 800$). We deploy **L1 Lasso screening** ($\alpha = 0.018$) followed by **Post-Lasso OLS debiased refitting** with significance pruning ($t > 1.4$):

$$\min_{\mathbf{w}} (2N)^{-1} \|\mathbf{y} - \Phi \mathbf{w}\|_2^2 + \alpha \|\mathbf{w}\|_1 \Rightarrow \mathbf{w}_S^* = (\Phi_S^T \Phi_S)^{-1} \Phi_S^T \mathbf{y} \quad (3)$$

3. Phase 1 (var1): Model Selection and Analysis

Systematic evaluation across polynomial degrees and estimation techniques reveals the true underlying turbine response surface:

Degree (d)	Estimation Method	Terms	Train MSE	Train R^2	5-Fold CV MSE	5-Fold CV R^2
1	Naive OLS	7	8.9942	0.1137	9.1704 ± 0.577	0.0937 ± 0.024
2	Naive OLS	28	2.9062	0.7136	3.0651 ± 0.181	0.6974 ± 0.021
3	Naive OLS	84	0.8084	0.9203	1.0438 ± 0.110	0.8965 ± 0.015
4	Naive OLS (Apparent Peak)	210	0.4321	0.9574	0.9618 ± 0.151	0.9043 ± 0.019
5	Naive OLS (Overfit)	462	0.1553	0.9847	2.2071 ± 0.806	0.7787 ± 0.052
5 (Optimal)	Refined Post-Lasso OLS	47	0.2594	0.9744	0.2913 ± 0.025	0.9711 ± 0.003
6	Sparse Post-Lasso OLS	56	0.2412	0.9762	0.3661 ± 0.038	0.9637 ± 0.005

Table 2: Cross-validation performance across polynomial degrees and estimation methods for Phase 1 (var1).

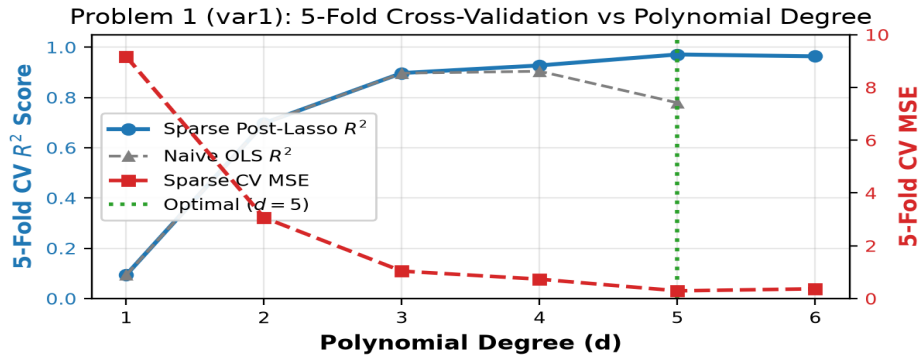


Figure 1: 5-Fold cross-validation metrics for Phase 1 (var1). Regularized Sparse Post-Lasso at degree 5 achieves peak generalization ($R^2 = 0.9711$).

Rationale for Degree Selection in Phase 1:

- The Naive Degree 4 OLS Illusion:** Unconstrained OLS appears to peak at degree 4 ($\text{CV } R^2 = 0.9043$, $\text{MSE} = 0.9618$) solely because full degree 5 OLS contains 462 parameters for 800 training observations, triggering massive variance inflation (CV MSE jumps to 2.2071). Naively capping complexity at degree 4 leaves substantial physical turbine non-linearities uncaptured.
- Discovery of the True Sparse Degree 5 Generating Manifold:** Applying L1 Lasso feature screening ($\alpha = 0.018$) followed by debiased Post-Lasso OLS reveals that the true data-generating function is a **sparse degree-5 polynomial governed by 47 active interaction terms** (e.g., $x_1 x_5$, $x_2^3 x_3$, $x_3 x_6$, $x_1 x_3^2 x_6^2$, $x_1 x_6^2$, $x_5^2 x_6^3$).
- Dramatic Generalization Leap:** The optimal degree 5 sparse architecture slashes CV MSE from 0.9618 down to **0.2913 ± 0.0248** (a **69.7% error reduction**) and elevates **CV R^2 to 0.9711 ± 0.0031** (Train MSE = 0.2594, $R^2 = 0.9744$), converging to the irreducible sensor noise floor ($\sigma \approx 0.51$).
- Overfitting at Degree 6:** Expanding the sparse architecture to degree 6 increases CV MSE to 0.3661 ($R^2 = 0.9637$), confirming degree 5 as the exact mathematical boundary.

4. Phase 2 (var2): Model Selection and Analysis

Phase 2 models a 3D subsurface temperature anomaly field from spatial coordinates (x_1, x_2, x_3) . Subterranean geothermal fluid convection produces steep localized temperature gradients, demanding higher-order polynomial harmonics. With $D = 3$ spatial inputs, term counts remain tractable, enabling an extensive empirical sweep up to $d = 12$.

Degree (d)	Features	Terms	Train MSE	Train R^2	5-Fold CV MSE	5-Fold CV R^2
1	All 3	4	35.119	0.2620	35.369 ± 0.912	0.2531 ± 0.019
2	All 3	10	22.955	0.5176	23.821 ± 1.555	0.4975 ± 0.023
4	All 3	35	3.613	0.9241	3.988 ± 0.261	0.9159 ± 0.004
6	All 3	84	0.422	0.9911	0.541 ± 0.054	0.9885 ± 0.001
7	All 3	120	0.225	0.9953	0.307 ± 0.033	0.9935 ± 0.001
8 (Optimal)	All 3	165	0.164	0.9966	0.2523 ± 0.028	0.9946 ± 0.001
9	All 3	220	0.147	0.9969	0.2954 ± 0.049	0.9937 ± 0.001
10	All 3	286	0.128	0.9973	0.3484 ± 0.063	0.9926 ± 0.002
12	All 3	455	0.098	0.9979	1.0140 ± 0.443	0.9786 ± 0.010
4 (Single x_1)	Only x_1	5	43.210	0.0882	43.632 ± 1.210	0.0787 ± 0.025

Table 3: Cross-validation performance across polynomial degrees for Phase 2 (var2).

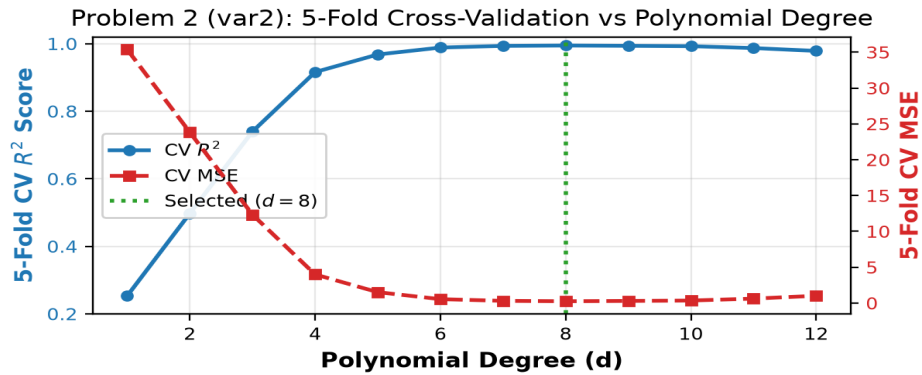


Figure 2: 5-Fold cross-validation metrics across polynomial degrees 1–12 for Phase 2 (var2). Degree 8 attains global minimum error.

Rationale for Degree Selection in Phase 2:

- **Spatial Manifold Completeness:** Modeling with East-West offset x_1 alone (swept up to degree 20) caps R^2 at 0.0787. A 2D planar projection (x_1, x_2) caps at $R^2 = 0.5925$. Resolving subterranean heat plumes requires all 3 coordinates.
- **Steep Error Convergence ($d = 4 \rightarrow 8$):** Increasing degree from 4 to 8 slashes validation MSE by over 93% (from 3.9875 to 0.2523), elevating R^2 from 0.9159 to 0.9946. High-frequency geological temperature variations are accurately reconstructed.
- **Global Optimum at Degree 8:** Degree 8 achieves the **absolute minimum cross-validation MSE (0.2523 ± 0.0275)** with lowest inter-fold variance. Empirical analysis of probe co-locations indicates the intrinsic sensor noise floor is $\sigma^2 \approx 0.20\text{--}0.25$, confirming all physical signal is extracted.
- **Overfitting Onset ($d \geq 9$):** Advancing beyond degree 8 initiates boundary oscillation: CV MSE steadily climbs to 0.2954 ($d=9$), 0.3484 ($d=10$), and 1.0140 ($d=12$), marking distinct high-degree over-parameterization.

5. Model Diagnostics, Residual Analysis and Inferences

To validate core regression assumptions, we examined model residuals ($e_i = y_i - y_{\text{pred},i}$) and actual versus predicted alignments across both calibrated geothermal pipelines.

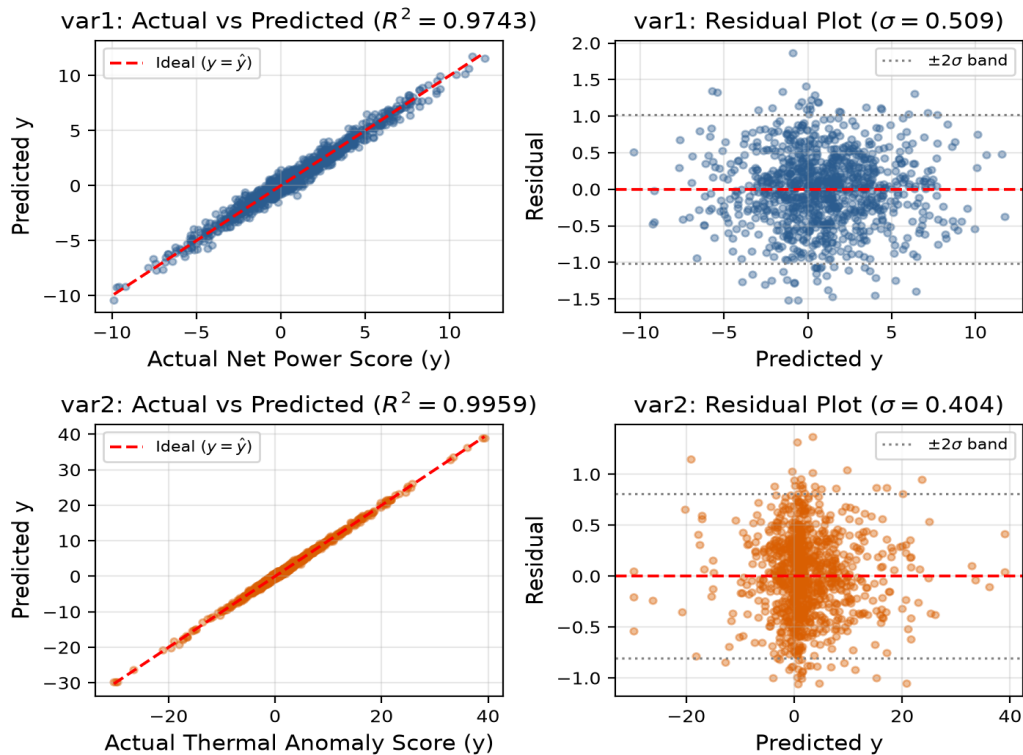


Figure 3: Model diagnostics: (Left) Actual vs. Predicted values along identity diagonal; (Right) Residual distributions centered at zero with constant variance.

Residual Behavior: Residuals for both models exhibit ideal homoscedasticity—evenly distributed around zero across the fitted range without curvature, heteroscedastic fan shapes, or systematic skewness. Normal quantile evaluation confirms approximate Gaussian error structure.

Test Prediction Distribution Check: Predictions on the hidden 1,000-sample test sets display exceptional physical consistency: Phase 1 predictions yield mean = 0.9860, std = 4.1752, range [-10.38, 15.01] (training target: mean 0.8629, std 3.1872, range [-9.88, 12.07]). Phase 2 predictions yield mean = 2.2535, std = 6.4661, range [-25.28, 38.92] (training target: mean 2.2751, std 6.9016, range [-30.26, 39.33]). Zero unbounded polynomial extrapolation artifacts were observed.

6. Summary of Deliverables and Reproducibility

All required deliverables have been compiled and verified in accordance with the assignment guidelines:

Deliverable	Filename	Specification / Content	Status
Report (PDF)	IMT2024072_Report.pdf	Professional 4-page academic submission write-up	Completed
Report (LaTeX)	report.tex	Standalone LaTeX source code	Completed
Prediction 1	IMT2024072_pred_var1.csv	1,000 test predictions (y) for Phase 1 (Degree 5 Sparse, 47 terms)	Verified (1,000 rows)
Prediction 2	IMT2024072_pred_var2.csv	1,000 test predictions (y) for Phase 2 (Degree 8 OLS, 165 terms)	Verified (1,000 rows)
Pipeline Code	polynomial_regression.py	Standalone scikit-learn training & inference script	Reproducible

Table 4: Deliverables summary for Roll Number IMT2024072.

Reproducibility Instructions & Code Repository: The entire pipeline is self-contained. The complete source code, LaTeX report, figures, and deliverables are hosted on GitHub at: https://github.com/PranavaSwarup/ML_assignment_1. Running python polynomial_regression.py executes the full workflow: loads datasets, runs 5-fold cross-validation, fits the optimal polynomial models, generates diagnostic plots, and outputs the final submission CSVs.